

|| Hindalco GET's CDP ||

|| Advanced Induction Program ||

2024

- ❖ **Academics:** Ph.D. in Mining Engineering from IIT(ISM) Dhanbad (2021)
 - M.Tech. in Mine Safety Engineering from AcSIR, CSIR-CIMFR, Dhanbad (2015) – 9.79 (A)
 - B.Tech. in Mining Engineering from IIT(BHU) Varanasi (2013) – 8.35 (absolute)
 - 10th and 12th from Kendriya Vidyalaya, Kankarbagh, Patna (2006 & 2008) – 86.4 & 84.4
- ❖ **Teaching Experience:** More than 2 Years
- ❖ **R&D Experience:** More than 9 Years as Scientist at CSIR-CIMFR, Dhanbad & NIRM, Bengaluru
- ❖ **Research Interest:** Applied Rock Mechanics, Underground Mining Methods, Numerical Simulation
- ❖ **Patents:** 2 **Citations:** 200+ **Foreign Visit:** USA & Czech Republic **Impact Factor:** 60+
- ❖ **Publications:** 43 [Journals-21 (Q1/Q2-8, SCI-15), Conferences-21, Book Chapter-1]
- ❖ **Ph.D. Guidance:** 12 (O) **M.Tech. Guidance:** 3 (C), **B.Tech. Guidance:** 12 (C)
- ❖ **Project Completed:** 22 [Inhouse-1, Foreign Collaborative-2, Sponsored-1 & Consultancy-18]
- ❖ **Ongoing R&D Project:** 3 [SAGES - 2 Cr, VRMS (Funded by Min. of Coal) -13 Cr, FRS (ISM) -15 Lacs]
- ❖ **Ongoing Consultancy Project:** 5



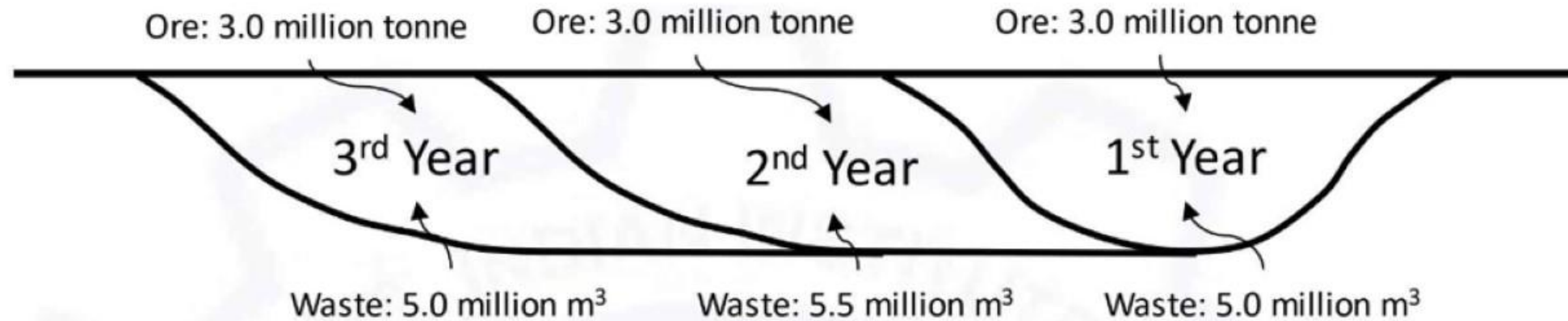
Topics to be covered

- ✓ **Mining, Mineral, and Sustainable Development**
- ✓ **Mine and Mineral Life Cycle**
- ✓ **Economic Geological Aspects In Exploration & Mining,**
- ✓ **Mineral Economics**
- ✓ **Classification of Mining Methods & Technology & Its Status in India & Developed Countries**

Question



Production planning of a small quarry having 3 years of life is shown in the figure. The following information of revenue and cost data are available.



Selling price of ore = Rs. 1500/tonne

Ore mining cost = Rs. 500/tonne

Waste mining cost = Rs. 500/m³

Initial capital = Rs. 1000 million

Discount rate = 10 %

By neglecting depreciation, salvage value and corporate tax, the NPV of the mining project, in million Rs., is _____. [round off to 2 decimal places]

Answer (35.50-38.30)

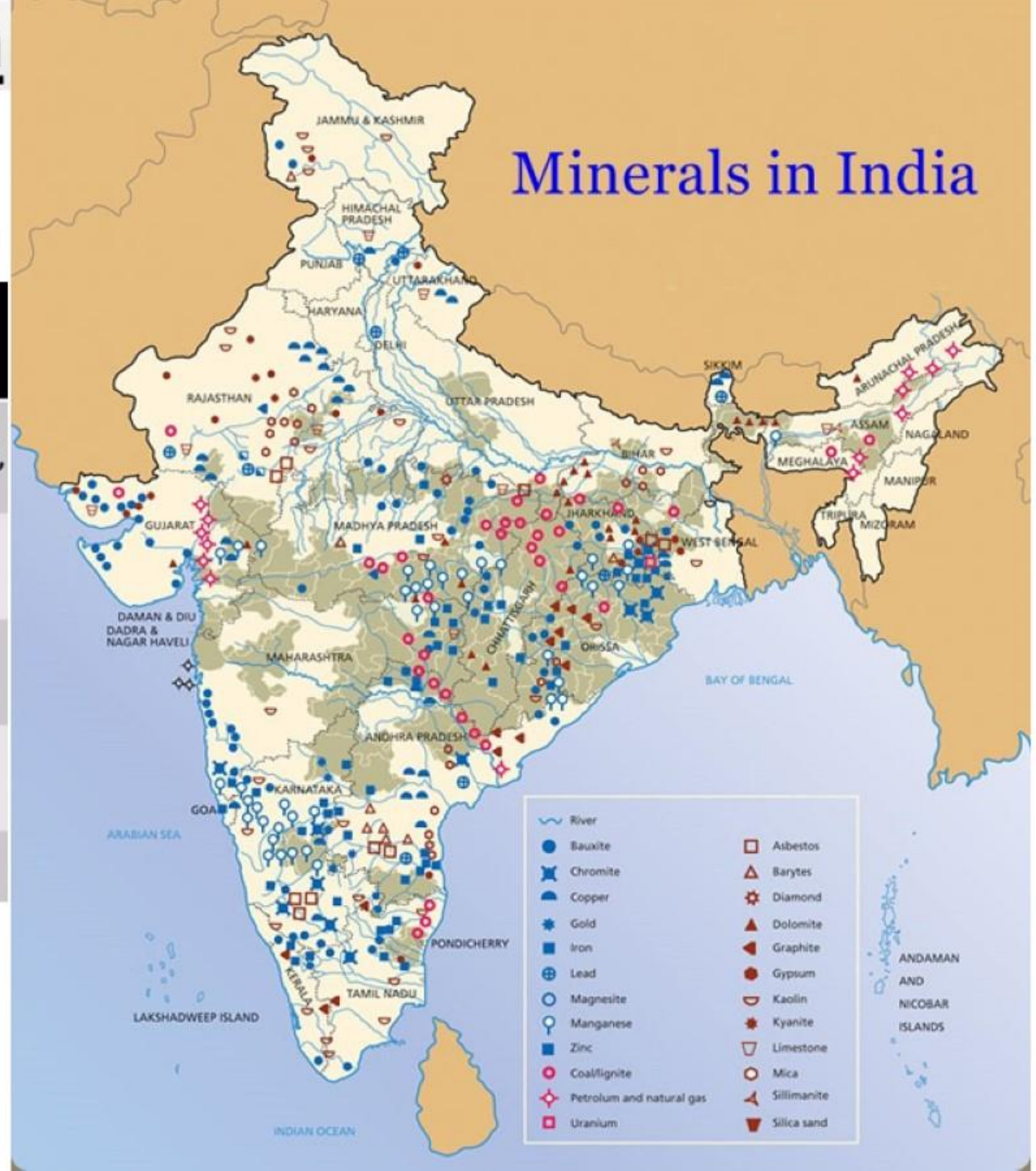
Solution: We know that,

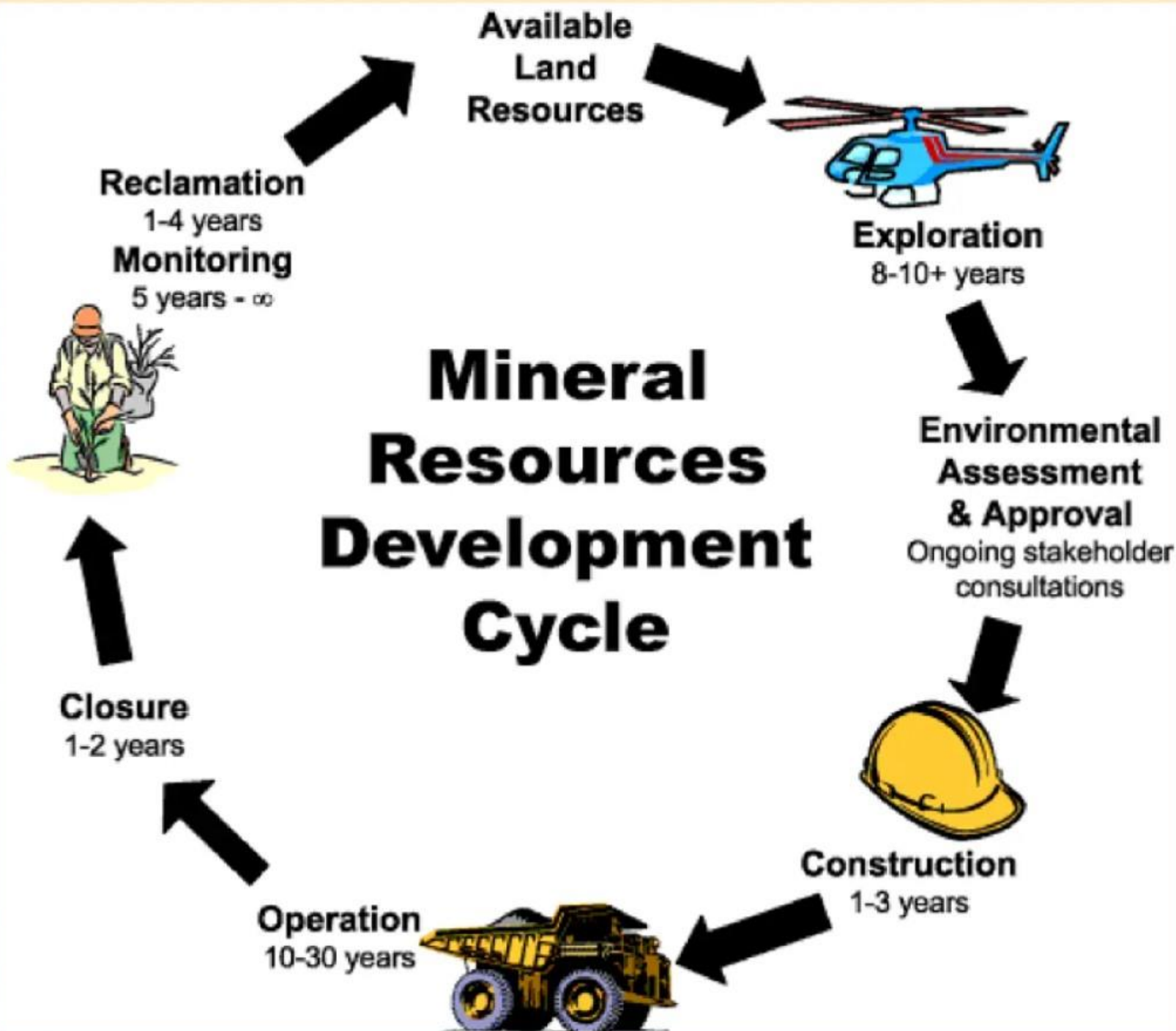
$$NPV = -1000 + \frac{(1500 - 500) \times 3 - (500 \times 5)}{1.1} + \frac{(1500 - 500) \times 3 - (500 \times 5.5)}{1.1^2} + \frac{(1500 - 500) \times 3 - (500 \times 5)}{1.1^3}$$

$$NPV = -1000 + 454.5454 + 206.6115 + 375.6574 = \text{Rs. } 36.8143$$

Geographical distribution

<u>Mineral Belt</u>	<u>Location</u>
North Eastern Peninsular Belt	Chota Nagpur plateau and the Orissa plateau covering the states of Jharkhand, West Bengal and Orissa.
Central Belt	Chattisgarh, Andhra Pradesh, Madhya Pradesh and Maharashtra
Southern Belt	Karnataka plateau and Tamil Nadu.
South Western Belt	Karnataka and Goa.
North Western Belt	Rajasthan and Gujarat along the Aravali Range.







MINING AND MINERAL RESOURCES OF INDIA



NONRENEWABLE MINERAL RESOURCES

▮ **Earth crust = Minerals + rock**

Minerals – inorganic compound that occurs naturally in the earth's crust

- ▮ Solid
- ▮ Regular internal crystalline structure.

▮ **Rock – solid combination of 1 or more minerals.**





- **Mineral Resource:** Any mineral useful to human kind
 - **Metallic Minerals:** Iron Oxide, Gold
 - Non-metallic mineral: Limestone, sand
 - **Fossil Fuel;** Coal, Petroleum
- **Ore:** A rock that can be **profitably** mined for a mineral (often a metal) or for minerals (metals)
 - **High Grade Ore;** has high concentration of the mineral
 - **Low Grade Ore:** smaller concentration
- **Gangue:** Minerals other than ore present in a rock



TYPES OF METALS

- **Abundant Metals:** concentration $>0.1\%$ in the earth's crust
 - Iron, Aluminum, Silicon, Magnesium, Titanium, Manganese
- **Scarce Metals:** $<0.01\%$ in the crust
 - **Ferro-alloys:** Nickel, Chromium
 - **Base Metals:** Copper, Lead, Zinc
 - **Precious or Noble Metals:** Gold, Silver, Platinum
 - **Special Metals:** Gallium, Arsenic, Germanium, Rare earths, Beryllium, Scandium etc..

- ▮ **PBT** = Persistent, Bio-accumulative and Toxic.
 - ▮ **5 nutrient metals:** Cu, Cr, Ni, Al, Zn
 - ▮ **6 non-nutrient metals:** Sb, As, Be, Cd, Pb, Hg
 (Antimony, Arsenic, Beryllium, Cadmium, Lead, Mercury)
- ▮ Metals cannot be banned and are present in nature, in soil, in food and in water
- ▮ Pb, Cd, As and Hg **pose special problem**

IMPORTANT TERMS



MINERAL RESOURCE: any naturally occurring non-living substance that is, has been, or may be useful to human beings

RESOURCE BASE: The volume of rock in a given area that contains more than the average concentration of that mineral in the Earth's crust.

INFERRED RESOURCES: Volumes of mineralization above some minimum grade that are known to exist, or are hypothesized on geological grounds to exist, in a given area, but have not been measured. May be orders of magnitude less than resource base and greater than the quantities of known ore.

INDICATED RESOURCES: mineral occurrences that have been sampled to a point where an estimate has been made, at a reasonable level of confidence, of their contained metal, grade, tonnage, shape, densities, physical characteristics. Generally orders of magnitude less than inferred resources in same area.

LOAD: An area in which there are specific indications of valuable mineralization.



PROSPECT: An area in which there are sufficient indications of the presence of ore to carry out a serious evaluation campaign (drilling, test mining). (It takes 10 – 100 **LEADS** to find 1 **PROSPECT**)

DEPOSIT: A defined or partially defined body of mineralization, which may or may not be ore, depending on economic conditions. (It takes 10 -100 **PROSPECTS** to develop 1 **DEPOSIT**)

GRADE: The percentage of the rock composed of valuable material. E.g. 62% Fe, 1% Cu,

ORE: Rock that can be mined for its mineral content at a profit.

ORE GRADE: Any grade above the lowest grade that can be mined at a profit. Varies with time and deposit.

RESERVES: Thoroughly explored and characterized volumes of ore (rarely amounts to more than a few years' extraction, because it costs a lot of money to convert resources to reserves). "Peak" people and many economists do not understand the difference between Resources and Reserves.

ORE GRADE for SELECTED ELEMENTS



Clarke: The average abundance of an element in the earth's crust

Clarke of Concentration:

The concentration of an element in a rock compared with its average concentration in the earth's crust.

Metal	Clarke (in percent)	Ore Grade (in percent)	Clarke of Concentration for Ore Grade
Aluminum	5.13	30	4
Iron - Fe	5.00	60	12
Titanium	0.66	15	23
Copper	0.0055	0.25	45
Rare Earths	0.019	1.6	84
Nickel	0.0075	1.5	200
Gold - Au	0.0000005	0.00023	460
Manganese	0.10	35	350
Uranium	0.0002	0.1	500
Zinc	0.007	4.0	600
Lead – Pb	0.0013	4.0	3000
Chromium	0.01	30	3000
Tin – Sn	0.0002	1.0	5000
Silver – Ag	0.00001	0.05	5000

In general, the larger the Clarke of Concentration, the smaller the typical orebody becomes

Ore Grade, or the Clarke of Concentration changes over time
and is affected by:

1. The availability of minerals with high concentrations of the element (e.g. Cu_2S , PbS , etc. REEs, for example, lack such minerals)
2. The difficulty and expense of recovering the metal from its ore (e.g. Aluminum, Titanium)
3. Advances in mining and refining technology
4. Changes in the value that society places upon the metal (e.g. Gold, REE, Lead, Tin etc..)



CLASSIFICATION OF ORE DEPOSITS

The purpose of classifying ore deposits into similar type's aids in the description of them, permits generalizations regarding Environments of formation and genesis, and improves the scientific ability to explore for, and discover them. To be most useful to a student, explorationist, and/or academic classifying something as complex and variable as ore deposits must be based on correct field and economic petrologic observations, as well as being as simple as possible.

Many attempts have been made to classify ore deposits since



Agricola's first efforts, but most attempts have been discarded because

- a) the field observations were not correct or were incomplete,
- b) the classification was too cumbersome, complex, and/or restrictive to have any practical value, and
- c) being based on insignificant or non useful variations between similar deposits.

Therefore to be practical and useful an ore deposit classification needs to be based on common descriptive factors followed by environments of formation; ideas on genesis can then follow these.

The descriptive factors need to be true and correct with environments of formation as complete and correct as the state of geological knowledge allows; this is also true for theories on the genesis of the deposit type. In other words the simpler and more field based the better and the hope is this is good enough to be an aid in the exploration and discovery of similar deposits.

The classification used here is a modification of Guilbert and Parks 1986 classification and is based on the field and geochemical description of an ore deposit along with the environment of formation. In classifying ore deposits in this manner it quickly becomes obvious that even though no two ore deposits are exactly alike, most of them fall into a relatively small number of categories.

Orthomagmatic

1. Cu-Ni Deposits

- a) Komatiite (ultramafic lava-hosted)
- b) Impact Melt
- c) Mafic intrusive rocks

2. PGE's

- a) Stratiform in mafic intrusions
- b) Mafic magmatic breccias

3. Diamond Deposits

- a) Kimberlites
- b) Lamprolites and lamprophyres

4. Chromite Deposits

- a) Layered intrusion-hosted
- b) Ophiolite-hosted
- 4a) Magnetite-apatite-vanadium deposits

5. Deposits Related to granites

- Pegmatites
- Uranium



Porphyry deposits are a type of mineral deposit that form from large-scale hydrothermal systems associated with intrusive igneous rocks.

They are characterized by the presence of porphyritic rocks that contain large crystals (phenocrysts) surrounded by a fine-grained matrix (groundmass).

The mineralization in porphyry deposits is typically associated with hydrothermal fluids that circulate through the porphyritic rocks, depositing minerals such as copper, gold, molybdenum, and silver in the form of sulfides and other minerals.

Hydrothermal Ore Deposits

6. Porphyry Cu-type deposits

- a) Porphyry copper in quartz diorite-granitic host rocks**
- b) Porphyry copper in tonolite to granodiorite host rocks**
- c) Porphyry copper in monzonite-syenodiorite host rocks**

7. Porphyry Molybdenum Deposits

- a) Porphyry Molybdenum in quartz-monzonite-granite host rocks**

8. Porphyry Gold Deposits

9. Porphyry Tin and Tungsten Deposits

10. Epithermal Gold Deposits

- a) Low sulfidation in felsic volcanic rocks**
- b) High sulfidation in felsic volcanic rocks**
- c) Sediment hosted deposits**

11. Orogenic Gold Deposits

- a) Mesothermal (mesozonal)-brittle-ductile)**
- b) Hypothermal (hypozonal)-(ductile)**

12. Volcanogenic Massive Sulfide Deposits

- a) Mafic volcanic rocks dominant**
 - ☐ **Ore hosted by felsic volcanic rocks**
 - ☐ **Ore hosted by mafic volcanic rocks**



□ Ore hosted by sedimentary rocks

b) Felsic volcanic rocks dominant

□ Ore hosted by volcanoclastic rocks

- low sulfidation deposits

- high sulfidation deposits

1) Ore hosted by felsic lava flows

2) high sulfidation deposits

3) low sulfidation deposits

13. Sedimentary Hosted Deposits

1. Mississippi Valley lead–zinc

2. Sedimentary-Exhalative

3. Evaporite Deposits-Salt and Potash

4. Sedimentary hosted copper

5. Sedimentary hosted uranium deposits

14) Iron Ore Deposits

a. Volcanic hosted

b. Sedimentary hosted (BIF)



15. Skarn Deposits

16. Iron Oxide-Copper-Gold Deposits

a. Calc Alkalic type

b. Alkalic type

17. Deposits Related to Surface Weathering

a) Mechanical Weathering-Placers

(including Witwatersrand)

b) Chemical Weathering

1) Bauxite

2) Nickel laterites

3) Manganese

4) Phosphates



INDIAN MINING SECTOR

- ✗ India is endowed with significant mineral resources. **India produces 95 minerals.**
- ✗ In India, 80% of mining is in coal and the balance 20% is in various metals and other raw materials such as gold, copper.
- ✗ **India ranks 2nd** in production of coal & lignite production.
- ✗ 11th in crude steel in the World.
- ✗ **Over 1.1 million people are employed in the Indian mining industry.** With over 2,326 private and 292 public operating mines in the country, minerals form 16 percent of India's exports.

MINE LIFECYCLE

1. Exploration Phase
2. Design Phase
3. Construction Phase
4. Production Phase
5. Post Closure Phase
6. Closure Phase



Exploration Phase

- It involve land issues
- bring an influx of a few job.
- some economic development through the purchase of local goods and services.

Design Phase

- This phase helps mining companies determine if and how a project can be safe, environmentally sound
- socially responsible, while meeting or exceeding regulatory requirements

Construction Phase

- This may take two years or more and involves building roads, processing facilities
- environmental management systems, employee housing and other facilities



Production

- Once construction is complete and systems have been tested, mineral production can begin
- It involves recovering, processing and transporting the minerals this process last depending on the size and type of the mineral deposit.

Post Closure Phase

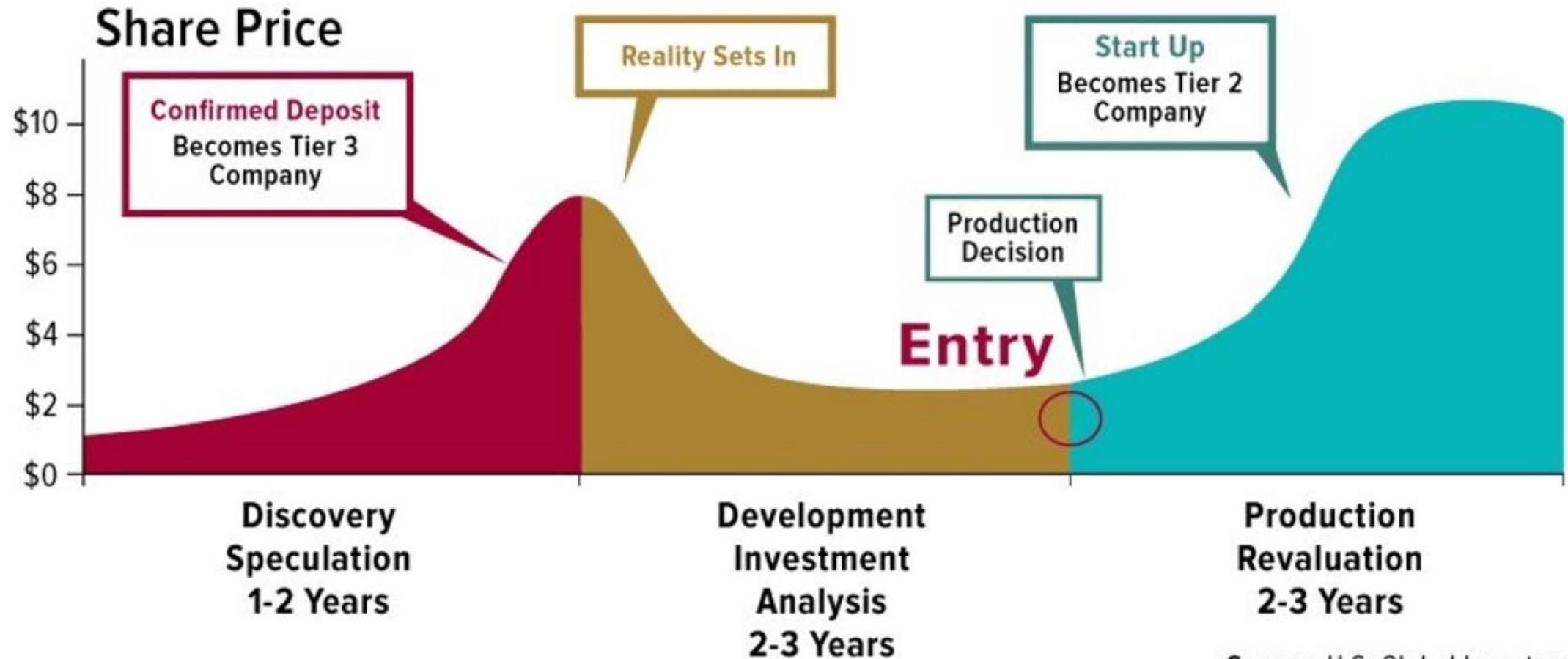
- Upon closing operations, measures must be taken to address environmental impacts that include water, vegetation and erosion..

Closure Phase

- a mine has a finite life span, and each site has a mine closure.
- includes dismantling the facilities and returning the mine site to a safe, stable and reclaimed state.
- Mine closure takes, on average, one to five years.



MINE LIFECYCLE



Source: U.S. Global Investors Research
USFUNDS.COM



MAJOR PLAYERS

1. **National Mineral Development Corporation**
Corporate office – Hyderabad, Andhra Pradesh.
 - ✓ A government of India organization and comes under the Ministry of steel, it is the largest iron ore producer in the country.
2. **Hindalco Industries**
Corporate office – Mumbai, Maharashtra .
 - ✓ Hindalco Industries is a company of Aditya Birla and one among the top 10 mining companies in India.

MAJOR PLAYERS

3. Gujarat Mineral Development Corporation Corporate office – Ahmadabad, Gujarat .

- ✓ Gujarat mineral development corporation limited is a leading mining company in India The organization is engaged in production of lignite, bauxite.

4. Vedanta Resource Corporate office – London, UK .

- ✓ Among the top mining companies in India, Vedanta resource is a metal and mining organization . The company is operating in many countries including India

COAL INDIA LIMITED



- ▮ Coal India Limited Government-owned entity which owns and operates India's mines. Under Ministry of Coal. *8 mining subsidiaries*, *1 mine siting* and *planning subsidiary*.
- ▮ *Coal India Limited* *calcutta* World's largest coal mining company.
- ▮ Responsible for approximately 85% of total Indian production. 324 million tons in 2005.
 - ▮ 1) Eastern Coalfields LTD
 - ▮ 2) Bharat Coking Coal LTD
 - ▮ 3) Central Coalfields LTD
 - ▮ 4) Central Mine Planning
 - ▮ 5) Northern Coalfields LTD
 - ▮ 6) South Eastern Coalfields LTD
 - ▮ 7) Western Coalfields LTD
 - ▮ 8) Mahanadi Coalfields LTD
 - ▮ 9) North Eastern Coalfields Source: Coal India LTD

THE MAJOR PLAYERS IN THE MINING SECTOR ARE CLASSIFIED ON THE BASIS OF THE MINERALS PRODUCED BY THEM NAMELY,:



- ▣ **Exploration and Production of Coal / Lignite :**
 - Coal India Ltd,
 - Neyveli Lignite Corporation,
 - IISCO(Indian Iron and Steel Company) etc.
- ▣ **Exploration of Metals (Copper, Bauxite, Iron Ore, Chromite, Lead- Zinc):**
 - NALCO ,
 - BALCO ,
 - Mineral Exploration Corporation Ltd,
 - Bharat Gold Mines Ltd,
 - ONGC,
 - Ircon ,
 - Hindustan Zinc Ltd ,
 - Hindustan Copper Ltd,
 - Sikkim Mining Corporation



▣ Iron Ore Sector:

- National Mineral Development Corporation
- Kudremukh Iron Ore Company,
- Steel Authority of India Ltd,
- Orissa Mining Corporation.

▣ Bauxite Mining and Aluminum Production:

- National Aluminum Company.

▣ Copper - Ore Mining:

- Hindustan Copper Ltd.

▣ Rock - Phosphate and Barytes Mining:

- Rajasthan State Mines and Minerals Ltd,
- Andhra Pradesh Mining Development Corporation.

GLOBAL MINING COMPANIES IN INDIAN MINING SECTOR:



- ▣ Transworld Garnet Co., Canada
- ▣ Meridian Peak Resources Corpn, Canada
- ▣ Pebble Creek Resources Ltd., Canada
- ▣ BHP Billiton, Australia
- ▣ Rio-Tinto Minerals Development Ltd., UK
- ▣ Metdist Group, UK
- ▣ Phelps Dodge Exploration Corpn , USA
- ▣ De-Beers Consolidated Mines ltd., South Africa
- ▣ Anglo American Exploration (India) BV,
Netherland

Environmental impact of the Mining industry





Land Use Management

Impact to
land and
surroundings

Waste
management

River water
pollution

Wildlife

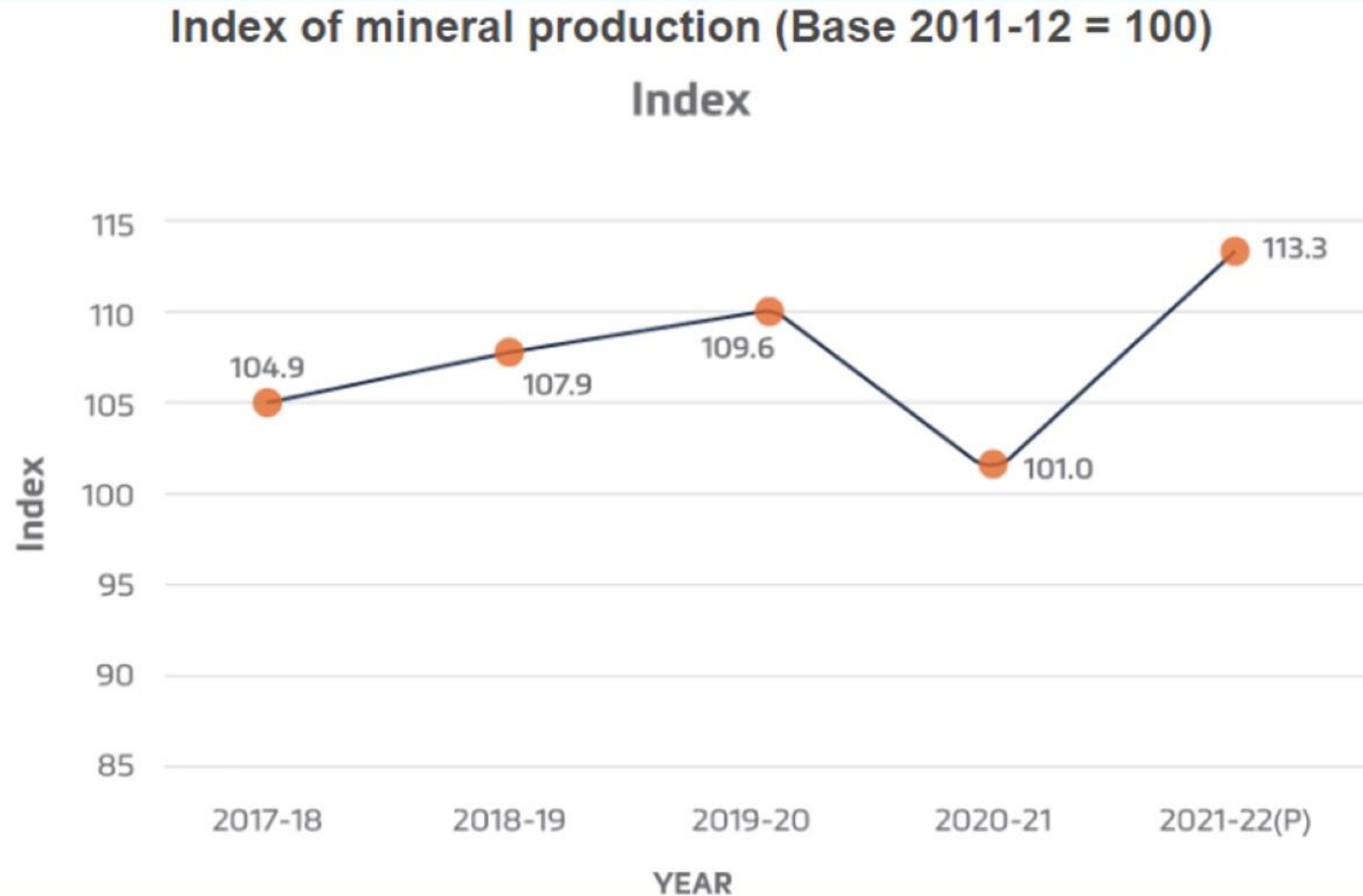


- ✓ Union minister urges mining sector to contribute 2.5% in GDP by 2026-27
- ✓ Minerals are valuable natural resources being finite and non-renewable.
- ✓ They constitute the vital raw materials for many basic industries and are a major resource for development.
- ✓ The history of mineral extraction in India dates back to the days of the Harappan civilization.
- ✓ The wide availability of the minerals in the form of abundant rich reserves made it very conducive for the growth and development of the mining sector in India.
- ✓ The country is endowed with huge resources of many metallic and non-metallic minerals.
- ✓ Mining sector is an important segment of the Indian economy.



- ✓ Since independence, there has been a pronounced growth in the mineral production both in terms of quantity and value.
- ✓ India produces as many as 95 minerals, which includes fuel, metallic, non-metallic, atomic and minor minerals (including building and other materials).

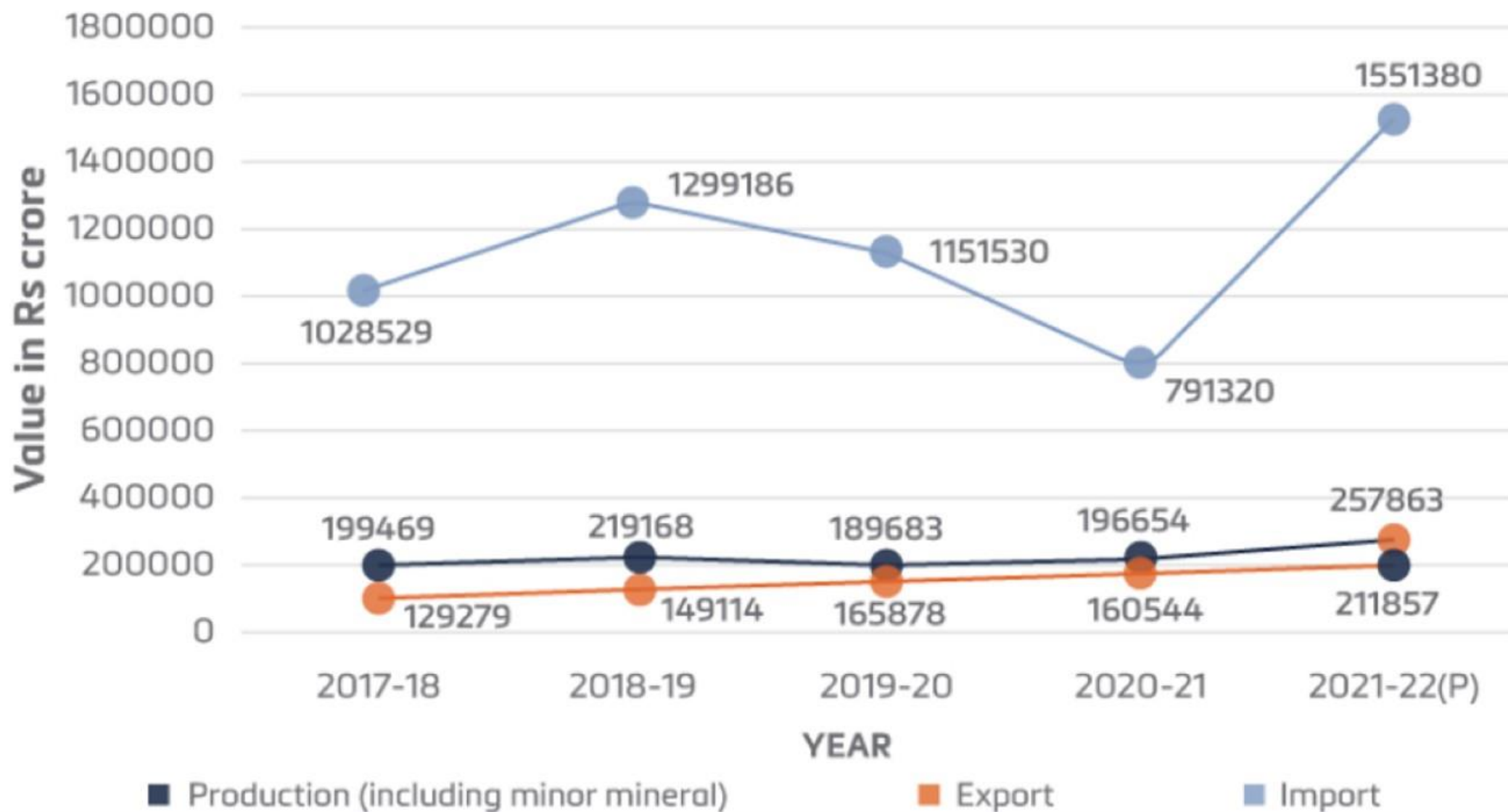
- ✓ Based on the overall trend so far the index of mineral production (base 2011-12) for the year 2021-22 is estimated to be 113.3 as compared to 101.0 for 2020-21 showing a positive growth of 12.17%.
- ✓ The decline in production is due to the restriction on exports, temporary discontinuance of mining for want of environmental clearance etc.



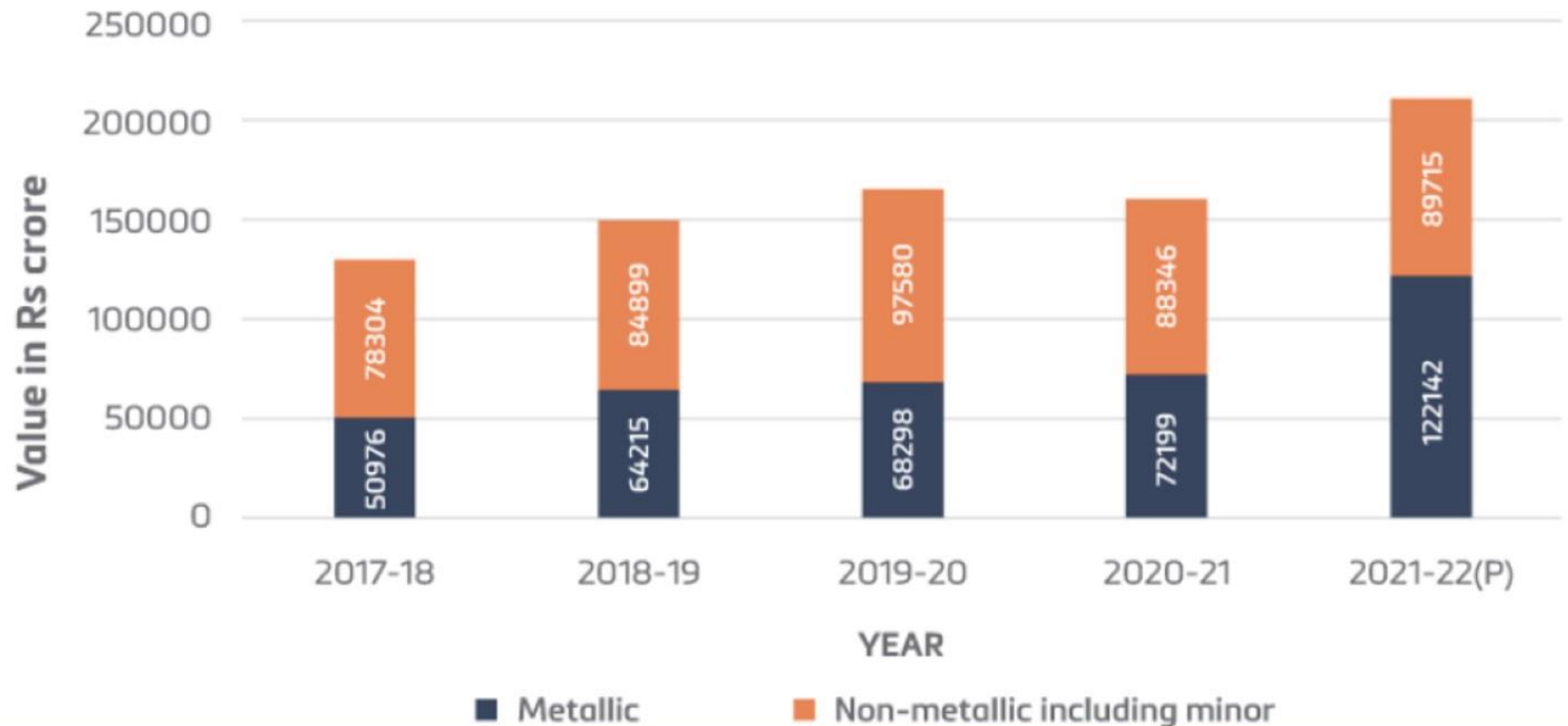


Mineral Production

Trends in Value of Mineral Production, Exports & Imports



Value of Minerals Production



- ✓ The total value of mineral production (excluding atomic and fuel minerals) during 2021-22 has been estimated at Rs. 211857 crore, which shows a increase of about 31.96% over that of the previous year.
- ✓ During 2021 -22, estimated value for metallic minerals, Rs 122142 crore or 57.65% of the total value and non- metallic minerals including minor minerals Rs 89715 crore or 42.35% of the total value.
- ✓ The WPI for minerals (base 2011-12=100) stood at 197.2 for the year 2021-22 and the corresponding index was 164.9 for the year 2020-21.
- ✓ The minerals included in the wholesale price index are bauxite, chromite, iron ore, copper conc., lead conc., garnet, zinc conc, manganese ore, limestone, phosphorite and sillamanite.

- ✓ The wholesale price index for metallic minerals was 193.3 in 2021-22 as compared to 159.8 in 2020-21 and that of other minerals was 211 in 2021-22 as compared to 183.1 in 2020-21.
- ✓ The Gross Value Added (GVA) accrued from mining and quarrying sector at 2011-12 prices is estimated by Central Statistical Office (CSO).
- ✓ The GVA (at current prices) for the year 2021-22, indicated that the mining and quarrying sector accounted for about 2.40% of GVA.
- ✓ The contribution of mining and quarrying sector to GVA for the year 2021-22 (at constant prices) is estimated at Rs. 327984 crore.
- ✓ This indicated an increase of 11.55% over that in the previous year.
- ✓ Similarly, the GVA (at current prices) for the year 2020-21, indicated that the mining and quarrying sector accounted for about 1.80% GDP.

- ✓ **The contribution of mining and quarrying sector to GVA for the year 2020-21(at constant prices) estimated at Rs. 294024 crore indicated a decrease of 8.62% over that in the previous year.**

MINING

- ✓ Indian mining industry is characterized by a large number of small operational mines.
- ✓ The number of mines which reported mineral production (excluding minor minerals, fuel minerals and atomic minerals) in India was 1319 in 2021-22 as against 1375 in the previous year.
- ✓ Out of 1319 reporting mines, 263 were located in Madhya Pradesh followed by Gujarat (147), Karnataka (132), Odisha (128), Chhattisgarh (114), Andhra Pradesh (108), Rajasthan (90), Tamil Nadu (88), Maharashtra (73), Jharkhand (45), and Telangana (39).
- ✓ These 11 States together accounted for 93% of total number of mines in the country in the year 2021 -22.

MINING

Number of Reporting Mines

Sector	2019-20	2020-21	2021-22(P)
Metallic Minerals	602	608	545
Non-Metallic Minerals	783	767	774
AllMinerals:Total*	1385	1375	1319

*Excluding atomic minerals, fuel mineral and minor minerals.

GROSS VALUE ADDED - MINING & QUARRYING SECTOR



MINING

Area wise Distribution of Mining Leases* (Frequency in Hect.)

As on 31.03.2021 (P) (All India)

Frequency Group Area in Hects.	Number of Mining Leases	Percentage of total leases	Area in "000 Hects.	Percentage of total area
0 to 2	387	12%	506.08	++
> 2to5	888	27%	3446.40	1%
> 5 to10	403	12%	2972.81	1%
>10to20	369	11%	5456.64	2%
>20to50	440	13%	14331.58	5%
>50to100	254	8%	18192.10	6%
>100to200	186	6%	26796.84	9%
>200to500	212	6%	69744.92	23%
Above500	175	5%	164951.39	54%
Total	3314	100	306398.76	100

MINING

The number of Underground Mines in operation Mineral wise (excluding fuel, atomic and minor minerals) (category based on leasehold area)

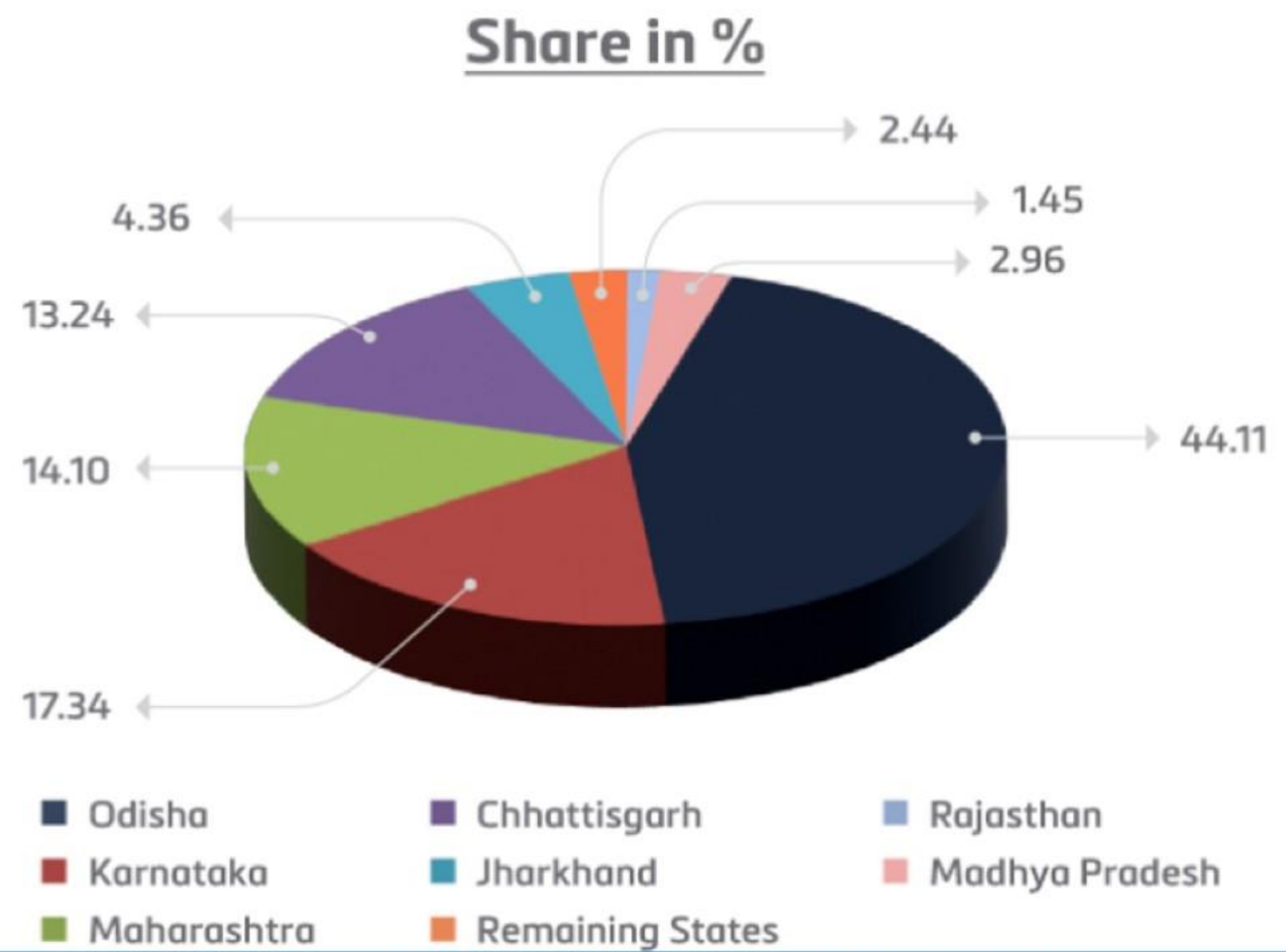
Mineral	Category A	Category B	Total
Chromite	2		2
Copper	8		8
Gold	8		8
Lead Zinc	10		10
Manganese	17		17
Rock Salt	1		1
Vermiculite		6	6
Total	46	6	52

MINING

- ✓ During the year 2021-22, mineral production was reported from 19 States of which the bulk of value of mineral production of about 97.04% was confined to 7 States only.
- ✓ The order was Odisha with a share of 44.11% followed by Chhattisgarh (17.34%), Rajasthan(14.10%),Karnataka (13.24%), Jharkhand (4.36%), Madhya Pradesh (2.44%), and Maharashtra (1.45%) in the total value of mineral production.
- ✓ Remaining 12 States have cumulative share of less than 3% of total value during the 2021-22.

MINING

Share of States in Value of Mineral Production 2021-22



MINING

- ✓ The States, which have indicated an increase in the value of mineral production, are Odisha (94.89%), Jharkhand (90.18%), Karnataka (62.22%), Chhattisgarh(60.88%), Maharashtra(59.30%), Madhya Pradesh(21.92%), Biha(21.59%), Andhra Pradesh(20.67%), Rajasthan(19.02%), Jammu & Kashmir(18.02%), Assam(14.45%), Telangana(14.29%), Himachal Pradesh(12.89%), TamilNadu(9.52%), Meghalaya (6.80%), Kerala (4.30%), Gujarat (3.88%) and Uttarakhand (2.57%).
- ✓ However, some of the principal mineral producing States revealed decrease in value of mineral production and those include Uttar Pradesh (-24.62%) and Goa (-100.00%).

MINING

- ✓ During the year 2021-22, the Private Sector contributed for 55.37% or Rs. 73502 crore (including metallic and non metallic) in the total value.
- ✓ The minerals which were wholly mined/ recovered by the private sector in 2021-22 were Garnet, Iolite, Lead conc, Limeshell, Siliceous earth, Sillimanite, Wollastonite, Zinc conc, Lead and zinc ore, Marl and Moulding sand.
- ✓ In 2021-22, the Private Sector accounted for sizeable 97.13% of Limestone, 77.43% of Vermiculite, 69.26% of Chromite, 60.70% of Iron Ore, 54.05% of Manganese Ore, 53.81% of Kyanite, and 50.80% of Bauxite
- ✓ During the year 2021-22, the Public Sector contributed for 44.63% or Rs. 59246 crore (including metallic and non metallic) in the total value.

MINING

- ✓ The minerals which were wholly mined / recovered by the public sector in 2021-22 were Copper ore and concentrate, Diamond, Fluorite (graded), Rock salt, Silver and Sulphur.
- ✓ In 2021-22, the Public Sector accounted for sizeable 99.08% of Gold Ore, 94.39% of Tin concentrate, 92.87% of Phosphorite., 90.75% of Selenite, 63.24% of Graphite and 54.03% of Magnesite.

MINING

- ✓ As per World Mineral Production, 2016-20, British Geological Survey, India's ranking in 2020 in world production in term of quantity was
- ✓ 2nd in coal & lignite, Steel (crude/liquid) and wollastonite, followed by
- ✓ 3rd in Zinc (slabs) and aluminium (primary);
- ✓ 4th in Chromite ores & concentrate, iron ore and graphite;
- ✓ 5th in Manganese ore;
- ✓ 6th in Bauxite;
- ✓ 7th in copper (refined);
- ✓ 10th in vermiculite;
- ✓ 15th in apatite & rock phosphate and
- ✓ 17th in magnesite.

GROSS VALUE ADDED - MINING & QUARRYING SECTOR



Contribution and Rank of India in World Production of Principal Minerals & Metals, 2020

MINING

Sector	Unit of Commodity	Production quantity		Contribution (Percentage)	India's rank in World order \$
		World	scope="col" India*		
Metallic Minerals					
Bauxite	'000tonnes	368600	20368	5.53	6 th
Chromite	'000tonnes	31000	2863	9.24	4 th
Ironore	milliontonnes	3016	204	6.76	4 th
Manganesecore	'000tonnes	49600	2688	5.42	5 th
Industrial Minerals**					
Magnesite	'000tonnes	28300	78	0.28	17 th
Apatite&Rock phosphate	'000tonnes	221000	1455	0.66	15 th
Metals					
Aluminium(primary)	'000tonnes	65400	3619	5.53	3 rd
Copper(refined)	'000tonnes	24900 ^{1/}	364 [#]	1.46	7 th
Steel(crude/liquid)	milliontonnes	1857	104	5.60	2 nd
Lead(refined)	'000tonnes	12500 ^{2/}	214 ^{#@}	1.71	3 rd
Zinc(slab)	'000tonnes	13800	715	5.18	3 rd



- ✓ India continued to be wholly or largely self-sufficient in minerals which constitute primary mineral raw materials that are supplied to industries, such as, iron & steel, aluminium, cement, various types of refractories, china clay based ceramics, glass.
- ✓ India is self-sufficient or near to self sufficient in bauxite, chromite, iron ore, kyanite, limestone, sillimanite etc. which are imported mainly for either blending with locally available mineral raw materials and/ or to meet special requirement for manufacturing special qualities of mineral based products.
- ✓ India is deficient in magnesite, manganese ore, rock phosphate, etc. which were imported to meet the domestic demand.



- ✓ To meet the increasing demand of uncut diamonds, emerald and other precious & semi-precious stones by the domestic Cutting and Polishing Industry, India is dependent on imports of raw uncut stones for their value-added re-exports.
- ✓ The degree of self-sufficiency in respect of various principal minerals and metals in 2020-21 is furnished

Self-Reliance in Minerals & Mineral Based Products



Degree of Self-sufficiency in Principal Minerals & Metals, 2020-21 (P)

Sl#.	Commodity	Apparent Consumption('000 tonnes)	Supply/Domestic supply ('000tonnes)	Order of self-sufficiency(%)
Minerals*				
1	Bauxite	23162	20369	88
2	Chromite	3017	2864	95
3	Ironore	147524	204481	100
4	Kyanite	5.9	4.9	83
5	Limestone	368439	349170 ^{1/}	95
6	Magnesite	437	78	18
7	Manganese ore	6664	2688	40
8	Rockphosphate (includingapatite)	9236	1456	16
9	Sillimanite	6.7	11.1	100
Metals				
10	Aluminium(primary)	2944	3619	100
11	Copper(refined)	720 ^{2/}	364	50
12	Lead(primary)	284 ^{3/}	214	76
13	Zinc	555 ^{4/}	715	100

Metallic Minerals

- ✓ **The value of metallic minerals in 2021-22 at Rs. 122142crore increased by about 69.18% over the previous year. Among the principal metallic minerals, iron ore contributed Rs 96381 crore or 78.91%, Zinc concentrate Rs. 8182crore or 6.70%, chromite Rs. 4730crore or 3.87%, Silver Rs 4212 or 3.45%, Bauxite Rs. 2477 crore or 2.03%, lead concentrate Rs. 2237 crore or 1.83%, manganese ore Rs. 2224 crore or 1.82%, copper (concentrate) Rs. 1095 crore or 0.90%, and Gold Rs. 601crore or 0.49%.**
- ✓ **The production of iron ore at about 253.97 million tonnes in 2021-22 registered an increase of 23.86% over the previous year.**

Metallic Minerals

- ✓ **About 39.30% of the total production was shared by Public Sector Companies like NMDC (16.07%), SAIL (13.31%), Odisha Mining Corporation (9.01%), etc.**
- ✓ **The share of Private Sector was 60.70% which included JSW Steel Ltd. (12.45%), Tata Steel (11.47%), Rungta Mines (5.56%), Vedanta Ltd (2.32%) and Arcelor Mittal India private Ltd (2.16%), etc.**
- ✓ **Almost the entire production of iron ore (98.62%) accrued from Odisha (53.82%), Chhattisgarh (16.27%), Karnataka (15.88%), Jharkhand (9.74%) and Madhya Pradesh (2.91%) during the year.**
- ✓ **The remaining 1.38% production was reported from Maharashtra, Rajasthan and Andhra Pradesh.**

Metallic Minerals

- ✓ **During the year 2021-22, the production of lead concentrate at 368 thousand tonnes decreased by 2.36% and that of zinc concentrate at 1594 thousand tonnes showed an increase of 5.29% over the previous year.**
- ✓ **Rajasthan accounted for the entire production of lead concentrate and zinc concentrate during the year 2021-22.**
- ✓ **The production of chromite at 3.79 million tonnes in 2021-22 increased by 33.75% as compared to the previous year.**
- ✓ **Odisha reported almost entire output of chromite (100%) in the country.**

Metallic Minerals

- ✓ **Private sector Chromite producers; viz, Tata Steel Mining Ltd. (47.28%), Indian Metals & Ferro Alloys Ltd. (14.78%), Ferro Alloys Corporation Ltd. (6.66%), Balasore Alloys Ltd (0.54%), cumulatively accounted for 69.26% of total production during 2021-22. Public Sector Company, Odisha Mining Corporation (OMC) reported 30.74% of the total production in 2021-22.**
- ✓ **The production of silver at 647140 kg in 2021-22 registered decrease of about 8.31% as compared to the previous year.**
- ✓ **Rajasthan was the leading producer of gold accounting for 99.98% of the total production.**
- ✓ **The remaining production was reported from Karnataka.**

Metallic Minerals

- ✓ **The production of bauxite at 22.49 million tonnes in 2021-22 decreased by 10.37% compared to the previous year.**
- ✓ **Four major companies, namely NALCO (33.12%), Utkal Alumina International Limited (26.67%), Odisha Mining Corporation (13.34%), HINDALCO (9.64%) dominated the total mining activities of bauxite of the country in 2021-22.**
- ✓ **Odisha accounted for 73.13% of the total output of bauxite during 2021-22 followed by Gujarat 8.97%, Jharkhand 8.04%, Chhattisgarh 4.30%, Maharashtra 2.85%, and Madhya Pradesh 2.71%.**
- ✓ **The production of manganese ore at 2.70 million tonnes in 2021-22 decreased by about 0.27% compared to that in the previous year.**

Metallic Minerals

- ✓ **MOIL continued to be the largest producer of manganese ore with a share of 45.79% of the total production in 2021-22 followed by, Sandur Manganese & Iron Ores Ltd (10.58%), Tata Steel (10.45%). Of the total production of manganese ore in 2021-22, Madhya Pradesh contributed 31.50%, Maharashtra 27.14%, Odisha contributed 19.01%, Karnataka 14.10%, Andhra Pradesh 7.57%.**
- ✓ **The production of copper concentrate at 114.42 thousand tonnes in 2021-22 increased by about 5.25% as compared to the previous year.**
- ✓ **Madhya Pradesh contributed 56.83% and Rajasthan 43.17% in total output.**
- ✓ **The production of primary gold at 1251 kg in 2021-22 registered increase of about 10.99% as compared to the previous year. Karnataka was the leading producer of gold accounting for 99.07% of the total production. The remaining production was reported from Jharkhand.**

Non-Metallic Minerals

- ✓ The value of production of non-metallic minerals at Rs 10606 crore during 2021-22 increased by 14.83% as compared to the previous year.
- ✓ Limestone retained its leading position by contributing 91.79% of the total value of non-metallic minerals in the year 2021-22.
- ✓ The other non-metallic minerals in the order of importance were phosphorite (7.18%), magnesite (0.42%), marl (0.31%), wollastonite (0.09%), graphite (0.09%), garnet (abrasive) (0.03%), siliceous earth (0.02%), diamond (0.02%), kyanite (0.02%), fluorite (0.01%), sillimanite, rock salt, moulding sand, vermiculite, selenite, limeshell, iolite.

Non-Metallic Minerals

- ✓ The production of limestone was at 393 million tonnes in the year 2021-22 increased by 12.50%, as compared to that in the previous year. Limestone is widely produced in India.
- ✓ As much as, 91.28% of the total output in the year 2018-19 was contributed by nine principal States; viz, Rajasthan (22.32%), Andhra Pradesh (12.80%), Madhya Pradesh (12.77%), Chhattisgarh (10.66%), Karnataka (10.03%), Telangana (7.26%), Gujarat (5.99%), Tamil Nadu (5.43%), and, Maharashtra (4.01%).
- ✓ The remaining 8.72% of the total production was shared by other limestone producing States.

Non-Metallic Minerals

- ✓ About 52.23% of total production was reported by principal producers, namely, Ultra Tech Cement Limited (17.76%), Shri Cement Limited (6.96%), Ambuja Cement (6.43%), ACC Limited (5.95%), Dalmia Cement (Bharat) Ltd. (4.24%), J.K. Cement Limited (3.46%), The Ramco Cements Limited (2.70%), JK Lakshmi Cement (2.39%), and Chettinad Cement corporation Ltd (2.36%).

Minor Minerals

- ✓ According to section 3(e) of the Mines and Minerals (Development and Regulation) Act, 1957 “Minor Minerals” means building stones, gravel, ordinary clay, ordinary sand other than sand used for prescribed purposes, and any other mineral which the Central Government may, by notification in the Official Gazette, declare to be a minor mineral. (For the purposes of this Act, the word "minerals" includes all minerals except mineral oils- natural gas and petroleum)
- ✓ Minor minerals account for over 55% of the total number of leases and nearly 60% of total leased area
- ✓ The value of production of minor minerals was estimated at Rs 79110 crore in the year 2020- 21.

Minor Minerals

- ✓ **Telangana with share of 21.79% in the value of minor minerals produced in the country occupied the top position.**
- ✓ **Rajasthan was at second place and had a share of 19.08% in the value of minor minerals.**
- ✓ **Next in the order was Andhra Pradesh with a share of 15.32%, Maharashtra 7.94%, Uttar Pradesh 7.10%, Gujarat 6.06%, Bihar 5.04%, Kerala 4.86%, Madhya Pradesh 4.68%, Karnataka 2.42%, West Bengal 2.31% and Chhattisgarh 1.31%.**
- ✓ **The cumulative contribution of remaining States and UTs was 1.72%.**

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Thank You

Queries??

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